# Amaury Pelzer

Boston, MA, USA | amaury.pelzer@gmail.com | (617) 415 8207 | amaurypelzer | amaurypelzer

## Profile

I am interested in using computational approaches to advance neuroscience research and healthcare. I am motivated by the potential of integrating genetics and machine/deep learning to uncover novel insights and improve clinical outcomes. I am especially passionate about working on clinical data.

## Skills

- Git (GitHub)
- Unix/Linux
- Bash
- HPC
- Slurm
- ML/DL
- Electronic Health Records
- Multi-omics
- Pathway Analysis
- Polygenic Risk Scores
- Python
- R
- C
- MATLAB
- SQL

## Education

**PhD** — **Maastricht University**, Psychiatric Genetics — Mar 2025 – Present
External PhD student at the Mental Health and Neuroscience Research Institute (Mhens)

**MSc** — **Maastricht University**, Cognitive and Clinical Neuroscience — Sep 2022 – Aug 2024
Thesis: *Multimodal Machine Learning for Outcome Prediction in Epilepsy Surgery*

**Minor** — **Vrije Universiteit Amsterdam**, Biomolecular Science and Neuroscience — Sep 2021 – Jan 2022

**BS** — **Maastricht University**, Biomedical Sciences — Sep 2019 – Aug 2022

## Experience

**Broad Institute of MIT and Harvard**, PhD student — Boston, USA, Mar 2025 – Present

**McLean Hospital (Harvard Medical School)**, Pre-Doctoral Research Fellow — Boston, USA, Mar 2025 – Present

**McLean Hospital (Harvard Medical School)**, Graduate Researcher — Remote, Oct 2024 – Mar 2025

- Integrating known haplotypes and imaging features with polygenic risk scores in UK Biobank PTSD and AD patients.

**The Florey (The University of Melbourne)**, Research Intern — Melbourne, Australia, Nov 2023 – Aug 2024

- Developed multimodal deep learning models for outcome prediction in epilepsy surgery by curating, cleaning, and preprocessing clinical datasets, creating machine learning prediction models using electronic health records (EHRs) in Python, and generating shell scripts to preprocess 3D T1-weighted MRIs. I also implemented deep learning models for prediction using pre- and post-surgical MRIs, created SLURM shell scripts for high-performance computing tasks, and developed an integrative approach combining clinical and imaging features to predict surgical outcomes.
- Conducted brain age prediction in epilepsy patients by performing SHAP analysis on machine learning and deep learning models, along with visualizing (in

both 2D and 3D) the brain regions most critical for brain age prediction, based on their Shapley values.

**Maastricht University**, Research Scientist — Maastricht, The Netherlands — Mar 2023 – Jul 2024

- Developed a KEGG-based pipeline for integrating metagenomics and transcriptomics data in the context of inflammatory bowel disease, utilizing both R and Python for data analysis and integration.
- Predicted mild cognitive impairment by investigating the discrepancy between genetic and actual educational attainment, which involved data cleaning and preprocessing of two clinical datasets, generating Polygenic Risk Scores for educational attainment, conducting survival analyses using Kaplan-Meier curves and Cox's proportional hazards regression, and performing statistical modeling with mixed models, all in R.

**Maastricht University**, Research Intern — Maastricht, The Netherlands — Mar 2022 – Jul 2022

- Modeled and analyzed pathways involved in iron metabolism disorders by creating and annotating a pathway in PathVisio, conducting network analysis using kinetic and transcriptomics data in Cytoscape, and identifying potential drug targets.
- Investigated the role of the microbiome in inflammatory bowel disease (IBD) through differential gene expression analysis using DESeq2, performing species-specific and community-level pathway analysis on metagenomics data in R, and identifying dysregulated microorganisms within the IBD microbiome.

## Conferences

**Machine Learning in Drug Discovery Symposium** — Nov 2024

**Precision Medicine 2024: Education in the AI Era** — Oct 2024

**Stanford AIMI Symposium** — May 2024

**Melbourne Epilepsy Meeting** — Mar 2024
Presented a poster titled: "*Machine Learning for Personalized Outcome Prediction following Epilepsy Surgery*".

**The Melbourne Brain Symposium** — Nov 2023

## Languages

- English (Fluent)
- French (Native)
- Japanese (Intermediate)
- Spanish (Basic)
- German (Basic)

## Certificates

Structuring Machine Learning Projects — Nov 2023

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization — Oct 2023

Neural Networks and Deep Learning — Jul 2023

Python for Genomic Data Science — Jan 2023